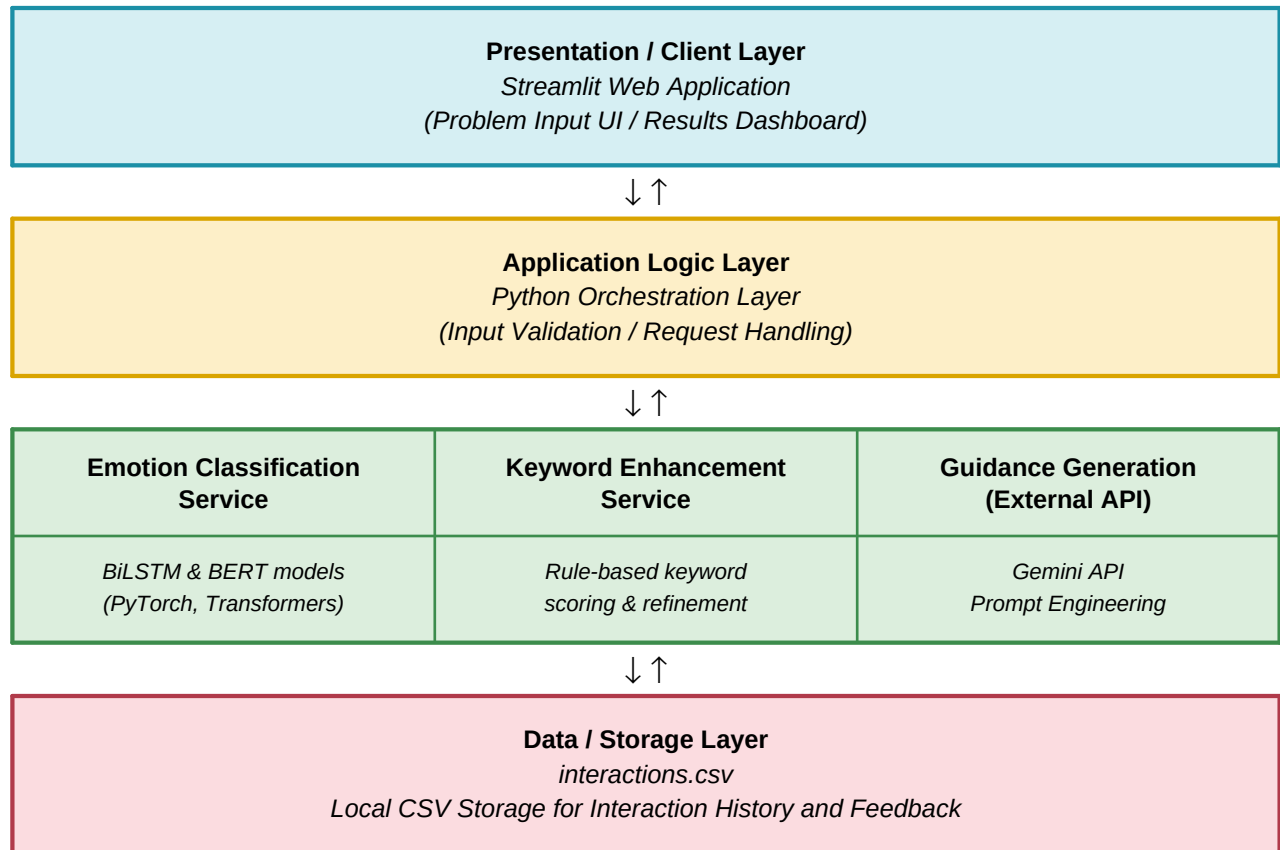


Solution Architecture

Date	28 June 2026
Team ID	—
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Provide a high-level visual representation of the solution's architecture, mapping the Presentation Layer, Application Layer, and Data Layer.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the UI for entering a problem description, viewing emotion results, guidance tips, and the analytics dashboard.	Streamlit, HTML, CSS
Application Logic Layer	Validates learner input, orchestrates model inference and Gemini API calls, and formats results for display.	Python
Emotion Classification Service	Predicts the learner's emotion (Bored, Confident, Confused, Curious, Frustrated) from free-text input.	PyTorch, BiLSTM, Hugging Face Transformers (BERT)

Component Name	Description / Role in Architecture	Technologies Used
Keyword Enhancement Service	Refines and boosts emotion predictions using rule-based keyword matching.	Python (rule-based logic)
Guidance Generation (External API)	Generates personalized tips, next steps, and encouragement using generative AI.	Gemini API, Prompt Engineering
Data / Storage Layer	Stores interaction history (input text, predictions, responses) for analytics and continuous learning.	CSV Files (interactions.csv)